HSDSD Midterm Study Notes

Audience: Student with EM fields and intro radio-wave background. **Source:** NTUST HSDSD lecture slides, chap1–5.

Chapter 1 — Why Interconnect Design Matters

The core problem

At high data rates, PCB traces can no longer be treated as simple wires. A signal with rise time $t_r = 100\,\text{ps}$ has a highest frequency of interest:

$$f_2 = \frac{1}{\pi t_r} = \frac{0.32}{t_r}$$

(chap1.pdf, p.6)

At 100 ps rise time, $f_2 \approx 3.2\,\text{GHz}$ — the corresponding wavelength in FR4 is only a few centimetres. Any trace longer than $\lambda/10$ must be treated as a **transmission line**, not a lumped wire.

Transmission line criterion

A conductor must be modelled as a TX line when its electrical length $\geq \lambda/10$:

$$\lambda_l = \frac{\lambda}{10} \quad \Leftrightarrow \quad TD_l \geq \frac{t_r}{10}$$

(chap1.pdf, p.7)

Practical rule: if trace delay $\geq t_r/10$, use TX-line model.

Trapezoidal spectrum

A clock waveform with rise time t_r, pulse width τ, and period T has a three-region spectral envelope:

Frequency range	Slope
0 to $f_1 = 1/(\pi\tau)$	0 dB/decade (flat)
f_1 to $f_2 = 1/(\pi t_r)$	-20 dB/decade
above f_2	-40 dB/decade

(chap1.pdf, p.6)

Key takeaway: faster edges raise f_2, increase spectral content, and worsen every interconnect effect.

Chapter 2 — Transmission Line Fundamentals

Characteristic impedance

For a general (lossy) line:

$$Z_o = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

(chap2.pdf, p.4)

For a lossless line ($R = G = 0$):

$$Z_o = \sqrt{\frac{L}{C}}$$

(chap2.pdf, p.4)

Practical impedance formulas

Microstrip (trace on outer layer, one ground plane):

$$Z_o \approx \frac{87}{\sqrt{\varepsilon_r + 1.41}} \ln\left(\frac{5.98H}{0.8W + T}\right)$$

(chap2.pdf, p.7)

Symmetric stripline (trace buried between two ground planes, equal spacing $H/2$ each side):

$$Z_o \approx \frac{60}{\sqrt{\varepsilon_r}} \ln\left(\frac{4H}{0.67\pi(T + 0.8W)}\right)$$

(chap2.pdf, p.8)

where H = total dielectric height, W = trace width, T = trace thickness.

Propagation velocity and delay

$$v = \frac{c}{\sqrt{\varepsilon_r}}, \qquad TD = \frac{x\sqrt{\varepsilon_r}}{c}$$

(chap2.pdf, p.10)

For microstrip the field is partly in air, so an **effective dielectric constant** $\varepsilon_{eff} < \varepsilon_r$ applies — microstrip is faster than stripline for the same ε_r.

SPICE modelling rule

Segment the line so each segment's time delay $\leq t_r/10$:

$$\text{segments} \geq \frac{10x}{vt_r}$$

(chap2.pdf, p.13)

Reflections

When a wave reaches an impedance discontinuity Z_t:

$$\rho = \frac{Z_t - Z_o}{Z_t + Z_o}, \qquad V_{\text{reflected}} = \rho\, V_{\text{incident}}$$

(chap2.pdf, p.18)

The first wave launched onto the line (voltage divider at source):

$$V_i = V_s \frac{Z_o}{Z_o + Z_s}$$

(chap2.pdf, p.17)

Use a **lattice diagram** to track successive reflections at each end of the line.

Chapter 3 — Crosstalk

Physical origin

Two adjacent traces couple through mutual inductance L_m and mutual capacitance C_m:

$$V_{\text{noise},L_m} = L_m \frac{dI_{\text{driver}}}{dt}, \qquad I_{\text{noise},C_m} = C_m \frac{dV_{\text{driver}}}{dt}$$

(chap3.pdf, p.4–5)

Inductive noise adds at the near end and subtracts at the far end. Capacitive noise adds at both ends (but in opposite polarity relative to inductive at far end).

NEXT and FEXT formulas (Case 1 — matched, long line)

Near-End CrossTalk (NEXT):

$$A = \frac{V_{\text{input}}}{4}\left(\frac{L_M}{L} + \frac{C_M}{C}\right)$$

(chap3.pdf, p.13)

NEXT is a **flat-top pulse** of duration $2\,TD$ (round-trip time), magnitude A.

Far-End CrossTalk (FEXT):

$$B = -\frac{V_{\text{input}}\, X\sqrt{LC}}{2T_r}\left(\frac{L_M}{L} - \frac{C_M}{C}\right)$$

(chap3.pdf, p.13)

FEXT is a **pulse** of duration T_r arriving at time TD. It is zero in a symmetric stripline (where $L_M/L = C_M/C$).

Mismatch correction

If the victim line is terminated with $R \neq Z_0$:

$$V_x = V_{\text{crosstalk}} \left(1 + \frac{R - Z_0}{R + Z_0}\right)$$

(chap3.pdf, p.15)

Coupling factor

$$K = \frac{L_{12}}{\sqrt{L_{11} L_{22}}}$$

(chap3.pdf, p.21)

K quantifies how tightly two lines are coupled (0 = no coupling, 1 = perfect coupling).

Odd- and Even-mode impedances

When two lines switch together in phase (even mode) or in opposite phase (odd mode):

Mode	Inductance	Capacitance	Impedance
Odd	$L_{11} - L_{12}$ (↓)	$C_{11} + C_m$ (↑)	$Z_{\text{odd}} < Z_0$
Even	$L_{11} + L_{12}$ (↑)	$C_{11} - C_m$ (↓)	$Z_{\text{even}} > Z_0$

(chap3.pdf, p.25–29)

- Odd mode: underdriven (source impedance too high) → slow settling
- Even mode: overdriven (source impedance too low) → ringing possible
- Stripline: $TD_{\text{odd}} = TD_{\text{even}}$ → no velocity variation
- Microstrip: $TD_{\text{odd}} \neq TD_{\text{even}}$ → different propagation speeds

SLEM — Single-Line Equivalent Model

For a three-line bus, line 2 effective impedance depends on its switching pattern relative to neighbours:

All in phase (common mode):

$$Z_{2,\text{eff}} = \sqrt{\frac{L_{22} + L_{12} + L_{23}}{C_{22} - C_{12} - C_{23}}}$$

(chap3.pdf, p.33)

Center out of phase with both neighbours (differential mode):

$$Z_{2,\text{eff}} = \sqrt{\frac{L_{22} - L_{12} - L_{23}}{C_{22} + C_{12} + C_{23}}}$$

These two modes produce the **worst-case** impedance extremes.

Termination for differential pairs

Pi-termination ($R_1 = R_2 = Z_{\text{even}}$ to ground, R_3 between lines):

$$R_3 = \frac{2Z_{\text{even}}Z_{\text{odd}}}{Z_{\text{even}} - Z_{\text{odd}}}$$

(chap3.pdf, p.39)

T-termination ($R_1 = R_2 = Z_{\text{odd}}$ in series, R_3 to ground):

$$R_3 = \frac{1}{2}(Z_{\text{even}} - Z_{\text{odd}})$$

(chap3.pdf, p.40)

Crosstalk minimisation rules

1. Widen spacing S between traces (most effective)
2. Minimise height H above ground plane for lower impedance $\rightarrow$ less coupling
3. Route on stripline to eliminate FEXT velocity effect
4. Keep parallel run lengths short
5. Route critical signals on orthogonal layers

Chapter 4 — Coupled Microstrip Line and Decoupling Techniques

Why FEXT exists in microstrip but not ideal stripline

In a microstrip, even-mode and odd-mode waves travel at **different speeds** because ε_{eff} is different for each mode:

$$T_d = TD_{even} - TD_{odd} > 0$$

The FEXT voltage is proportional to this time difference:

$$|V_4(TD_{even})| = V_{max} \cdot \min\left(\frac{T_d}{t_r}, 1\right)$$

(chap4.pdf, p.14)

$$V_{max} = \frac{Z_0}{Z_{even} + 2Z_0 + Z_{odd}} V_s$$

(chap4.pdf, p.14)

In an ideal stripline $T_d = 0$, so FEXT $= 0$.

Even- and odd-mode time delays

$$TD_{even} = l\sqrt{(L_s + L_m)C_s}$$

(chap4.pdf, p.10)

$$TD_{odd} = l\sqrt{(L_s - L_m)(C_s + 2C_m)}$$

(chap4.pdf, p.10)

Front-end decoupling capacitor C_f

Place a capacitor C_f between the two lines at the source end. This adds $2C_f/l$ per unit length to the odd-mode circuit only, slowing the odd mode to match the even mode:

$$TD^c_{odd} = l\sqrt{\left(C_s + 2C_m + \frac{2C_f}{l}\right)(L_s - L_m)}$$

(chap4.pdf, p.23)

Design goal: choose C_f so $TD^c_{odd} = TD_{even}$ ($T_d = 0$).

Trade-off: a lumped C_f reduces FEXT but **increases NEXT** (the capacitor charging current appears as a near-end spike) and degrades eye diagram at high bit rates.

RC time constant of the C_f circuit

$$\tau = 2R_tC_f, \qquad R_t = \frac{Z_0 Z_{odd}}{Z_0 + Z_{odd}}$$

(chap4.pdf, p.28)

Distributed decoupling capacitors

Distributing C_f as N smaller capacitors spaced $\Delta l = \lambda/10$ apart:

- Eliminates FEXT (≈ 0)
- Reduces NEXT (no large spike — gradual charging)
- Restores $|S_{11}|$ and eye diagram performance
- Much better than lumped C_f at high frequencies

The optimal spacing is at or below $\lambda/10$ at the signal's 3-dB bandwidth:

$$f_{3\text{dB}} = \frac{0.35}{t_r}, \qquad \lambda = \frac{3 \times 10^8}{f_{3\text{dB}}\sqrt{\varepsilon_{eff}}}$$

(chap4.pdf, p.39)

Chapter 5 — Nonideal Interconnect Issues

DC resistance

$$R = \frac{\rho L}{Wt} \quad \Omega$$

(chap5.pdf, p.4)

Significant only for long lines, narrow traces, or multiload buses.

Skin effect

At high frequencies, current crowds to the conductor surface. The skin depth is:

$$\delta = \sqrt{\frac{2\rho}{\omega\mu}} = \sqrt{\frac{\rho}{\pi F \mu}} \quad \text{(meters)}$$

(chap5.pdf, p.6)

63% of current flows within one skin depth δ.

Resistance increases as $\sqrt{F}$ above the transition frequency F_t (where $\delta = t$).

AC resistance — microstrip signal conductor

$$R_{\text{ac signal}} \approx \frac{\rho}{W\delta} = \frac{\sqrt{\rho\pi\mu F}}{W} \quad \Omega/\text{m}$$

(chap5.pdf, p.9)

Ground return AC resistance

The ground return current follows a Lorentzian distribution. 79.5% is within $\pm 3H$, so effective ground width $\approx 6H$:

$$R_{\text{ac ground}} \approx \frac{\sqrt{\rho\pi\mu F}}{6H} \quad \Omega/\text{m}$$

(chap5.pdf, p.11)

Total microstrip AC resistance

$$R_{\text{ac microstrip}} \approx \sqrt{\rho\pi\mu F}\left(\frac{1}{W} + \frac{1}{6H}\right) \quad \Omega/\text{m}$$

(chap5.pdf, p.12)

SPICE combined resistance

$$R_{\text{total}} \approx \sqrt{R_{\text{ac}}^2 + R_{\text{dc}}^2}$$

(chap5.pdf, p.10)

Surface resistance

$$R_s = \frac{L}{W}\sqrt{\rho\pi\mu} \quad \text{ohms} \cdot \sqrt{\text{s}}$$

(chap5.pdf, p.16)

so that $R_{\text{ac}} = R_s\sqrt{F}$. *(chap5.pdf, p.15)*

Stripline AC resistance

Parallel combination of the microstrip resistance seen from the top ground plane (H_1) and the bottom ground plane (H_2):

$$R_{\text{ac stripline}} = \frac{R_{(H1)} \cdot R_{(H2)}}{R_{(H1)} + R_{(H2)}} \quad \Omega/\text{m}$$

(chap5.pdf, p.13)

Dielectric losses

The dielectric constant becomes complex under a time-varying field:

$$\varepsilon = \varepsilon' - j\varepsilon''$$

(chap5.pdf, p.20)

Loss tangent (easier to look up in data sheets than ε''):

$$\tan\delta_d = \frac{\varepsilon''}{\varepsilon'}$$

(chap5.pdf, p.21)

This adds a **shunt conductance** per unit length to the TX-line model:

$$G = \frac{\varepsilon''}{\varepsilon'}(2\pi F C_{11}) \quad \text{S/m}$$

(chap5.pdf, p.21)

Important: For FR4 up to 15 GHz, dielectric loss dominates over conductor loss. *(chap5.pdf, p.5)*

FR4 dielectric constant variation

FR4 is not a constant-ε_r material. First-order model:

$$\varepsilon_r = \varepsilon_{\text{rsn}} V_{\text{rsn}} + \varepsilon_{\text{gls}} V_{\text{gls}}$$

(chap5.pdf, p.29)

Empirical formula (frequency in kHz):

$$\varepsilon_r(V_{\text{rsn}}, F) = 6.32 - [2.17 + 0.168 \log_{10} F\,(\text{kHz})]\, V_{\text{rsn}}$$

(chap5.pdf, p.29)

ε_r decreases with frequency (4.7 at low MHz $\rightarrow$ 3.9 at 1 GHz for typical FR4).

Serpentine traces

Due to coupling between adjacent parallel sections, part of the signal shortcuts through crosstalk (Path 2) and arrives **early**:

- Duration of "ledge" $\propto$ parallel section length L_p
- Ledge voltage $\propto$ spacing S

Rules: $S \geq 3H$ to $4H$; minimise L_p; do not serpentine clock traces. *(chap5.pdf, p.33)*

Intersymbol Interference (ISI)

ISI occurs when reflections or noise from a previous bit have not settled before the next transition. Worst when:

$$T_{\text{period}} < 2 \times TD_{\text{line}}$$

Rules: minimise reflections; keep traces short; avoid tightly coupled serpentines; minimise crosstalk. *(chap5.pdf, p.38)*

90° bends

Excess capacitance at a corner (area ≈ 0.054 squares):

$$C_{90\text{ř bend}} \approx C_{11}\, w$$

(chap5.pdf, p.39)

Mitigation: use 45° bends or chamfer the corner.

Topology effects

- **Balanced T** (equal leg lengths, equal loading): clean step response.
- **Unbalanced T** ($L_1 \neq L_2$): severe ringing from multiple reflections bouncing between junction and receivers.

Rules: symmetric topology (equal length and loading on every branch); minimise discontinuity at the junction. *(chap5.pdf, p.44)*

Quick Reference — Key Formulas

Concept	Formula	Source
Highest frequency	$f_2 = 0.32/t_r$	chap1.pdf, p.6
TX line criterion	trace delay $\geq t_r/10$	chap1.pdf, p.7
Lossless Z_o	$\sqrt{L/C}$	chap2.pdf, p.4

Concept	Formula	Source
Microstrip Z_o	$\frac{87}{\sqrt{\varepsilon_r+1.41}} \ln \frac{5.98H}{0.8W+T}$	chap2.pdf, p.7
Stripline Z_o	$\frac{60}{\sqrt{\varepsilon_r}} \ln \frac{4H}{0.67\pi(T+0.8W)}$	chap2.pdf, p.8
Propagation delay	$TD = x\sqrt{\varepsilon_r}/c$	chap2.pdf, p.10
Reflection coeff.	$\rho = (Z_t - Z_o)/(Z_t + Z_o)$	chap2.pdf, p.18
NEXT (matched)	$A = \frac{V}{4}(L_M/L + C_M/C)$	chap3.pdf, p.13
FEXT (matched)	$B = -\frac{VX\sqrt{LC}}{2T_r}(L_M/L - C_M/C)$	chap3.pdf, p.13
Odd-mode Z	$\sqrt{(L_{11} - L_{12})/(C_{11} + C_m)}$	chap3.pdf, p.27
Even-mode Z	$\sqrt{(L_{11} + L_{12})/(C_{11} - C_m)}$	chap3.pdf, p.29
Skin depth	$\delta = \sqrt{\rho/(\pi F \mu)}$	chap5.pdf, p.6
AC resistance	$R_s\sqrt{F}$	chap5.pdf, p.15
Loss tangent	$\tan \delta_d = \varepsilon''/\varepsilon'$	chap5.pdf, p.21
Dielectric conductance	$G = (\varepsilon''/\varepsilon')(2\pi F C_{11})$	chap5.pdf, p.21